

Some Descriptive Set Theory

Polish Spaces and Standard Borel Spaces

Definition: A topological space X is called a *Polish space* if it is separable and completely metrizable.

Theorem: Let X be an uncountable Polish space. Then, X has a closed subset Y which is a Cantor set.

Proof. Let d be a compatible metric on X . Let Z be the subset of X consisting of all points x such that each neighborhood of x is uncountable. Then, Z is closed.

For each $x \in X \setminus Z$, let r_x denote the supremum of all $r \in (0, 1)$ such that the open ball centered at x with radius r is countable.

Let S be a countable dense subset of $X \setminus Z$. Then, the open balls around $x \in S$ with radius $\frac{1}{2}r_x$ for each $x \in S$ cover $X \setminus Z$. In particular, this means that $X \setminus Z$ is countable. Therefore, we may assume that $Z = X$.

By a recursive procedure, we may construct pairwise disjoint closed balls $B_{n,1}, B_{n,2}, \dots, B_{n,2^n}$ such that each $B_{n,k}$ has a positive radius no larger than $1/n$ and contains two of the sets $B_{n+1,1}, B_{n+1,2}, \dots, B_{n+1,2^{n+1}}$. Then, we have that the set

$$C = \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{2^n} B_{n,k}$$

is a closed subset of X that is a Cantor set. □

Definition: A measurable space $(X, \mathcal{S})$ is called a *standard Borel space* if it is isomorphic as measurable spaces to $(Y, \mathbf{B}(Y))$, where $\mathbf{B}(Y)$ denotes the Borel σ -algebra, for some Polish space Y . Equivalently, there is a Polish topology $\mathcal{T}$ on X such that $\mathcal{S} = \mathbf{B}(\mathcal{T})$.

A notable fact about Borel sets in Polish spaces is that we can consider Polish topologies that have a base of clopen sets. This follows from two lemmas.

Lemma: If $(X, \mathcal{T})$ is Polish, and $F \subseteq X$ is closed, then if $\mathcal{T}_F$ is the topology generated by $\mathcal{T} \cup \{F\}$, then $\mathcal{T}_F$ is Polish, F is clopen in $\mathcal{T}_F$, and $\mathbf{B}(\mathcal{T}_F) = \mathbf{B}(\mathcal{T})$.

Lemma: Let $(X, \mathcal{T})$ be a Polish space, and let $(\mathcal{T}_n)_{n \in \mathbb{N}}$ be a sequence of Polish topologies on X with $\mathcal{T} \subseteq \mathcal{T}_n$ for each n . Then, the topology $\mathcal{T}_\infty$ generated by the union of all the $\mathcal{T}_n$ is Polish.

Moreover, if $\mathcal{T}_n \subseteq \mathbf{B}(\mathcal{T})$ for each n , then $\mathbf{B}(\mathcal{T}_\infty) = \mathbf{B}(\mathcal{T})$.

Proof. Let $X_n = X$ be the sequence of Polish spaces with topology $\mathcal{T}_n$. Then, consider the map $\varphi: X \rightarrow \prod_n X_n$ given by the diagonal embedding $x \mapsto (x, x, \dots)$.

Then, we have that $\varphi(X)$ is closed in the product topology on $\prod_n X_n$, for if $(x_n)_n \notin \varphi(X)$, then for some $i < j$, we have $x_i \neq x_j$, so there are disjoint open subsets in $\mathcal{T}$ (hence open in $\mathcal{T}_i$ and $\mathcal{T}_j$) such that $x_i \in U$, $x_j \in V$, and $(x_n)_n \in \pi_i^{-1}(U) \cap \pi_j^{-1}(V)$, where π_i denotes the projection map onto coordinate i .

Therefore, $\varphi(X)$ is Polish, but since φ is a homeomorphism of $(X, \mathcal{T}_\infty)$ with $\varphi(X)$, it follows that $(X, \mathcal{T}_\infty)$ is Polish.

Furthermore, if $\mathcal{T}_n \subseteq \mathbf{B}(\mathcal{T})$, and $\{U_i^{(n)}\}_{i \in \mathbb{N}}$ is a basis for $\mathcal{T}_n$, then $\{U_i^{(n)}\}_{i, n \in \mathbb{N}}$ is a subbasis for $\mathcal{T}_\infty$, so $\mathcal{T}_\infty \subseteq \mathbf{B}(\mathcal{T})$. □

Theorem: Let $(X, \mathcal{T})$ be a Polish space, and let $A \subseteq X$ be a Borel set. Then, there is a Polish topology $\mathcal{T}_A \supseteq \mathcal{T}$ such that $\mathbf{B}(\mathcal{T}_A) = \mathbf{B}(\mathcal{T})$ and A is clopen in $\mathcal{T}_A$.

Proof. Consider the class $\mathcal{S}$ of subsets A of X for which there exists a Polish topology $\mathcal{T}_A \supseteq \mathcal{T}$ with $\mathbf{B}(\mathcal{T}_A) = \mathbf{B}(\mathcal{T})$ and A is clopen in $\mathcal{T}_A$.

It is enough to show that $\mathcal{T} \subseteq \mathcal{S}$ and $\mathcal{S}$ is a σ -algebra. We observe that $\mathcal{T} \subseteq \mathcal{S}$ from the first lemma and that $\mathcal{S}$ is closed under complements. Finally, if $\{A_n\}_{n \in \mathbb{N}} \subseteq \mathcal{S}$ is some countable collection of subsets, then if $\mathcal{T}_n := \mathcal{T}_{A_n}$ satisfy the condition for A_n , we may define $\mathcal{T}_\infty$ as above, and see that $A = \bigcup_{n \in \mathbb{N}} A_n$ is open in

$\mathcal{T}_\infty$, so we may consider the topology $\mathcal{T}_A$. In particular, this gives that $\mathcal{S}$ is closed under countable unions. $\square$

Countable Borel Relations, Invariant Measures, and Graphings

Equivalence Relation Preliminaries

First, we collect some definitions.

Definition:

- A relation R on a set X is a set of ordered pairs $R \subseteq X^2$. We will write xRy to denote $(x, y) \in R$. We also define the *sections*

$$R_x = \{y : (x, y) \in R\}$$

$$R^y = \{x : (x, y) \in R\}$$

and the inverse relation

$$R^{-1} = \{(y, x) : (x, y) \in R\}.$$

- If X is equipped with a Borel structure, then the relation R (i.e., equivalence relations, partial orders, total orders, etc.) is called Borel if $R \subseteq X^2$ is Borel.
- We will let $\text{dom}(f)$, $\text{ran}(f)$, and $\text{graph}(f)$ denote the domain, range, and graph of f respectively.
- A graph G with vertex set X is a non-reflexive, symmetric relation on X . The neighbors of $x \in X$ in the graph G are the set of all $y \in X$ such that $(x, y) \in G$. The cardinality of the set of neighbors of x is called the degree of x , denoted $d_G(x)$.
- A path in G is a finite sequence of vertices $(x_0, x_1, \dots, x_n)$ where $(x_i, x_{i+1}) \in G$, and $x_i \neq x_j$ whenever $i \neq j$ except for possibly $x_n = x_0$. The equivalence classes of the relation xEy if and only if there exists a path from x to y defines the connected components of the graph. If a graph has one connected component, we say the graph is connected.
- If $x_n = x_0$ with $n \geq 3$, then we call the path a cycle. A graph that has no cycles is called acyclic, and a connected acyclic graph is called a tree. A rooted tree is a tree with a distinguished vertex x_0 called its root.
- If X is a standard Borel space and E is a Borel equivalence relation on X , then we say the relation is countable if $[x]_E$ is countable for each $x \in X$.
- If $\Gamma \curvearrowright X$ is a countable group acting on X such that Γ preserves the Borel structure of X , then we define the orbit equivalence relation to be $xE_\Gamma^X y$ if and only if there exists g such that $g \cdot x = y$.
- The orbit space X/Γ is defined to be the quotient X/E_Γ^X .
- If E is a countable Borel relation, then the *full group* of E , denoted $[E]$, is the set of Borel automorphisms whose graphs are contained in E . The set of partial Borel automorphisms is denoted $[[E]]$; that is,

$$[[E]] = \{\phi : A \rightarrow B : \phi \text{ is a bijection, } A, B \subseteq X \text{ are Borel, } \text{graph}(\phi) \subseteq E\}.$$

- We call a countable equivalence relation E on X *aperiodic* if every equivalence class $[x]_E$ is infinite.
- Given Borel equivalence relations E and F on X and Y respectively, we say E is *Borel reducible* to F , denoted $E \leq_B F$, if there is a Borel map $f : X \rightarrow Y$ such that xEy if and only if $f(x)Ff(y)$.

Example: If X^Γ denote the space of functions of Γ onto X , then the left shift action of Γ on X^Γ is defined by $\gamma \cdot p(\delta) = p(\gamma^{-1}\delta)$.

If G is a standard Borel group and $\Gamma \subseteq G$ is a countable discrete subgroup (also known as a lattice), then the action of Γ onto G by left translation, taking $\gamma \cdot g = \gamma g$, yields the quotient space G/Γ given by the set of *right* cosets of Γ .

Example: The following are examples of countable Borel equivalence relations.

- (i) If $X = 2^{\mathbb{N}}$, then the *eventual equality* relation says that $x E_0 y$ if and only if there exists m such that for all $n \geq m$, we have $x_n = y_n$. The *tail equivalence* relation says that $x E_t y$ if and only if there exist m and k such that for all n , we have $x_{m+n} = y_{k+n}$.
- (ii) If $X = \mathbb{R}$, then the *Vitali relation* says that $x E_v y$ if and only if $x - y \in \mathbb{Q}$.
- (iii) If $X = \mathbb{R}_+$, then the *commensurability relation* says that $x E_c y$ if and only if $x/y \in \mathbb{Q}$.
- (iv) If $k \geq 2$ and X is the space of subshifts of $k^{\mathbb{Z}}$, where a subshift is a nonempty closed subset of $k^{\mathbb{Z}}$ invariant under the shift map $(S(x))_i = x_{i-1}$.

This is a compact subspace of the hyperspace of compact subsets of $k^{\mathbb{Z}}$, so it is a compact and metrizable space. Define two subshifts to be isomorphic if there is a homeomorphism between the two closed subsets that commutes with the shift, and define an equivalence relation as such.

Theorem (Feldman–Moore): Let E be a countable Borel equivalence relation on X . Then, there is a countable group Γ and a Borel action of Γ on X such that $E_\Gamma^X = E$.

Proposition (Marker Lemma): Let E be an aperiodic countable Borel equivalence relation on X . Then, there is a sequence $\{S_n\}_{n \geq 1}$ of Borel sets $S_n \subseteq X$ such that

- (i) $S_0 \supseteq S_1 \supseteq S_2 \supseteq \dots$;
- (ii) $\bigcap_{n \geq 1} S_n = \emptyset$;
- (iii) every S_n is a complete section for E (i.e., it has nonempty intersection with each equivalence class of E).

Such a sequence is called a *vanishing sequence of markers*.

Proof. Via some results from descriptive set theory, we may assume that the equivalence relation E lives on the set $X = 2^{\mathbb{N}}$. Then, we let $s_n(x)$ be the lexicographically smallest $s \in 2^n$ such that $|[x]_E \cap \mathcal{N}_s| = \infty$, where $\mathcal{N}_s$ denotes the cylinder set in X corresponding to s .

Define $x \in A_n$ if and only if $x|_n = s_n(x)$. Then, we have that $\{A_n\}_{n \geq 1}$ is a decreasing sequence of Borel sets which intersects each class of E at infinitely many points. Then, $A = \bigcap_{n \in \mathbb{N}} A_n$ intersects each class of E in at most one point, so that $B_n = A_n \setminus A$ is the vanishing sequence of markers as desired. $\square$

Invariant Measures

Definition: If E is a countable Borel equivalence relation on X , then we say μ is E -invariant if for all $f \in [E]$, we have $f_*\mu = \mu$, where $f_*\mu$ is the pushforward measure, defined by

$$f_*\mu(A) = \mu(f^{-1}(A)).$$

Proposition: The following are equivalent:

- (a) μ is E -invariant;
- (b) μ is Γ -invariant for any countable group Γ such that $E = E_\Gamma^X$;
- (c) μ is Γ -invariant for some Γ such that $E = E_\Gamma^X$;
- (d) for all $\phi \in [[E]]$, we have $\mu(\text{dom}(\phi)) = \mu(\text{ran}(\phi))$.

If μ is invariant under E , then we may define a measure $\bar{\mu}$ on E by taking

$$\bar{\mu}(A) = \int |A_x| d\mu(x),$$

where again,

$$A_x = \{y : (x, y) \in A\}.$$

It is also possible to define

$$\bar{\mu}'(A) = \int |A^y| d\mu(y),$$

but since μ is invariant, $\bar{\mu} = \bar{\mu}'$. This follows from the fact that

$$E = \bigcup_{i \in \mathbb{N}} \text{graph}(g_i)$$

for some collection of relation-preserving automorphisms $g_i \in [E]$. If $A \subseteq E$ is any Borel subset, then

$$A = \bigcup_{i \in \mathbb{N}} (\text{graph}(g_i) \cap A),$$

so there are $f_i \in [[E]]$ such that $\text{graph}(f_i) = \text{graph}(g_i) \cap A$. Therefore, we may write any $A \subseteq E$ as a countable disjoint union of graphs of functions $f \in [[E]]$. Therefore, it suffices to show that $\bar{\mu}(\text{graph}(f)) = \bar{\mu}'(\text{graph}(f))$ for any $f \in [[E]]$. If $C = \text{dom}(f)$ and $D = \text{ran}(f)$, then $\bar{\mu}(\text{graph}(f)) = \mu(C)$ and $\bar{\mu}'(\text{graph}(f)) = \mu(D)$, so by the definition of $[[E]]$, we are done.

Now, we may define an equivalence relation $\tilde{E}$ on E by taking $(x, y) \tilde{E} (z, w)$ if and only if $x E z$, or equivalently, $x E y E z E w$.

Proposition: The invariant measure M on E is $\tilde{E}$ -invariant.

Proof. Let Γ act on X such that $E = E_\Gamma^X$. Then, if we define the action $(g, h) \cdot (x, y) = (g \cdot x, h \cdot y)$, then we have that $E_{\Gamma^2}^E = \tilde{E}$, so it suffices to show that the action preserves M .

For this, if $A \subseteq E$ is Borel, we have

$$M((g, h) \cdot A) = \int |\{(g \cdot x, h \cdot y) : (x, y) \in A\}| d\mu(x).$$

Setting $B = (g, h) \cdot A$, then we have $y \in B_x$ if and only if $(x, y) \in B$, which holds if and only if $(g^{-1} \cdot x, h^{-1} \cdot y) \in A$, which holds if and only if $h^{-1} \cdot y \in A_{g^{-1} \cdot x}$. Therefore, $B_x = h \cdot A_{g^{-1} \cdot x}$, so that

$$\begin{aligned} M((g, h) \cdot A) &= \int |A_{g^{-1} \cdot x}| d\mu(x) \\ &= \int |A_x| d\mu(x) \\ &= M(A), \end{aligned}$$

where we use the fact that μ is invariant under Γ . □

Graphings and Cost

Definition: If $E \subseteq X^2$ is a countable Borel equivalence relation, then a (Borel) *graphing* of E is a graph G such that the connected components of G are the equivalence classes of E . We say that G generates E .

Definition: A Levitt graph (L-graph) is a countable family $[[E]] \supseteq \Phi = \{\varphi_i\}_{i \in I}$ of partial Borel isomorphisms $\varphi_i: A_i \rightarrow B_i$, where A_i, B_i are Borel subsets of X .

We call Φ *finite* if I is finite. We say Φ is an Levitt graphing (L-graphing) if Φ generates E in the sense that $x E y$ if and only if $x = y$ or there is a finite set $i_1, \dots, i_k \in I$ and $\varepsilon_1, \dots, \varepsilon_k \in \{1, -1\}$ such that $x = \varphi_{i_1}^{\varepsilon_1} \cdots \varphi_{i_k}^{\varepsilon_k}(y)$.

For every L-graph Φ , there is an associated graph G_Φ that generates the same equivalence relation. We say $xG_\Phi y$ if and only if $x \neq y$ and there exists i such that $\varphi_i(x) = y$ or $\varphi_i(y) = x$.

Conversely, for every graph G , there is an L-graph Φ_G such that $G = G_{\Phi_G}$. This follows from fixing a Borel (total) ordering $<$ on X , and finding a family of partial Borel functions $\{f_i\}_{i \in \mathbb{N}}$ with disjoint graphs such that for all $x \in X$, we have

$$\{(x, y) : xGy \text{ and } x < y\} = \bigcup_{i \in \mathbb{N}} \text{graph}(f_i).$$

Since the f_i are countable-to-one, we have $\{g_{i,j}\}_{j \in K_i}$ a family of partial Borel automorphisms with disjoint graphs such that

$$\text{graph}(f_i)^{-1} = \bigcup_{j \in K_i} \text{graph}(g_{i,j})$$

(Recall that we define the inverse of a relation to be the coordinates flipped.) Define $\Phi_G = \{g_{i,j}\}_{i,j}$.

Considering the case of an E -invariant μ , if G is a graphing of $E|_A$ for some conull E -invariant Borel subset A , then we say G is a graphing of E almost everywhere. Similarly, we may define an L-graphing almost everywhere.

Now, we will discuss the theory of cost.

Definition: Let E be a countable Borel equivalence relation on X , and let μ be an E -invariant measure with corresponding measure on E given by M . If $G \subseteq E$ is a subset, then we define its *cost* to be

$$C_\mu(G) = \frac{1}{2}M(G).$$

Observe that $0 \leq C_\mu(G) \leq \infty$, and $0 < C_\mu(G)$ if G is a graphing of E provided that it's not the case that almost every E -class is a singleton. This follows from the fact that G is a complete section for the relation $\widetilde{E}$ on E , meaning that $M(G) > 0$.

Now, if $\Phi = \{\varphi_i\}_{i \in I}$ is an L-graph, define its cost by taking

$$C_\mu(\Phi) = \sum_{i \in I} \mu(\text{dom}(\varphi_i)) = \sum_{i \in I} \mu(\text{ran}(\varphi_i)).$$

Then, we have

$$C_\mu(\Phi) = \frac{1}{2} \int \sum_{i \in I} (\chi_{A_i}(x) + \chi_{B_i}(x)) d\mu(x),$$

where we have $A_i = \text{dom}(\varphi_i)$ and $B_i = \text{ran}(\varphi_i)$. If we have the graph G_Φ is the graph associated to Φ , then

$$|(G_\Phi)_x| \leq \sum_{i \in I} (\chi_{A_i}(x) + \chi_{B_i}(x)),$$

so $C_\mu(G_\Phi) = \frac{1}{2}M(G_\Phi) \leq C_\mu(\Phi)$.

Meanwhile, if $G \subseteq E$ is a graph with associated L-graph Φ_G , then we notice that $(x, y) \mapsto (y, x)$ is in the full group $[\widetilde{E}]$ of $\widetilde{E}$. Therefore, we have (for a fixed Borel total ordering $<$)

$$\begin{aligned} C_\mu(G) &= \frac{1}{2}M(G) \\ &= M(\{(x, y) : xGy \text{ and } x < y\}) \\ &= M\left(\bigcup_{i \in \mathbb{N}} \text{graph}(f_i)\right) \end{aligned}$$

$$\begin{aligned}
&= \sum_{i \in \mathbb{N}} M(\text{graph}(f_i)) \\
&= \sum_{i \in \mathbb{N}} M(\text{graph}(f_i)^{-1}) \\
&= \sum_{i \in \mathbb{N}} \sum_{j \in K_i} M(\text{graph}(g_{i,j})) \\
&= C_\mu(\Phi_G).
\end{aligned}$$

Therefore, we may define the *cost* of the equivalence relation E as

$$\begin{aligned}
C_\mu(E) &= \inf \{ C_\mu(G) : G \text{ is a graphing of } E \} \\
&= \inf \{ C_\mu(\Phi) : \Phi \text{ is an L-graphing of } E \}.
\end{aligned}$$

Observe that $0 \leq C_\mu(E) \leq \infty$, and $C_\mu(\Delta_X) = 0$, where Δ_X is the equality relation on X .

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